Can Vision-Language Models Assess Human Skill from Video?

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Abstract

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Chapter 1

Background

This chapter sets out the background needed to interpret the experiments that follow. It begins with the problem of assessing skill from video and the supervised tradition that has addressed it, then describes the dataset that supplies the benchmark used throughout, then covers the architecture of the models evaluated, the limitations already documented for them, and finally the prompting strategies from which the four prompt formats of Chapter 3 are drawn.

1.1 Skill assessment from video

Skill is not a visual category in the way that "climbing" is. Two clips can contain the same action, the same equipment and the same scene composition, and differ only in the efficiency, economy and precision of the movement. The evidence that separates a novice from an expert is distributed across time and across fine kinematic detail: a hesitation before committing to a hold, an over-gripped hand, an unnecessary weight shift, a leg that swings where it should step. This is what makes skill assessment different from action recognition. Recognising what is being done is a categorical judgement supported by object and scene evidence available in almost any single frame; judging how well it is being done is an evaluative judgement over a continuous quality that must be integrated over the whole performance.

Psychological accounts of expertise supply the conceptual scaffolding for the label scale used in this work. The Dreyfus model of skill acquisition (Dreyfus and Dreyfus, 1980) describes a staged progression from novice through advanced beginner and competent to proficient and expert, in which the qualitative character of the performance — not merely its speed or its error rate — changes at each stage. Ericsson's deliberate-practice framework (Ericsson et al., 1993) similarly treats expertise as a continuum accumulated over long training horizons. Both accounts imply that a coarse binary novice/expert split discards the structure that makes expertise interesting, which motivates benchmarks with more than two levels. Both also predict that adjacent levels are genuinely hard to separate, a prediction the results of Chapter 4 confirm emphatically.

The computational lineage of this problem is action quality assessment (AQA). Early work regressed judges' scores in Olympic diving and gymnastics from spatiotemporal features; later systems adopted deep architectures with multi-task objectives that predict action class and quality jointly (Parmar and Morris, 2019). Because quality labels are inherently noisy, uncertainty-aware score distribution learning (Tang et al., 2020) models them as distributions rather than point values. Fine-grained benchmarks such as FineDiving (Xu et al., 2022) push towards procedure-aware assessment, breaking an action into sub-steps that are scored individually.

A parallel line addresses skill determination specifically: deciding which of two people performs a task better, rather than assigning an absolute score. Doughty et al. (2018) framed this as pairwise deep ranking, learning from relative rather than absolute supervision, and later extended it with rank-aware temporal attention for long videos (Doughty et al., 2019), on the observation that only a fraction of a long video is actually diagnostic of skill.

Two properties of this literature matter for the present work. First, it is uniformly supervised and uniformly domain-specific: a model trained to rank surgical suturing does not transfer to bouldering, and building a system for a new activity requires a new annotated dataset, which is expensive precisely because the annotation must come from a domain expert rather than a crowd worker. Second, and important for interpreting the results reported here, trained methods on the EgoExo4D proficiency benchmark report roughly 47–53% top-1 accuracy on the four-class task (Grauman et al., 2024). The task is therefore solvable, at roughly twice chance, by a model that has been fitted to it. Any zero-shot failure reported in Chapter 4 is consequently a transfer failure, and not evidence that the labels are noise.

1.2 The EgoExo4D dataset

EgoExo4D (Grauman et al., 2024) is a large multi-view dataset of skilled human activity, captured simultaneously from an Aria headset worn by the participant (egocentric) and from four static cameras placed around the scene (exocentric). Each recording is a take: one participant performing one activity. The dataset spans physical activities such as bouldering, basketball, dance and soccer, and procedural activities such as cooking and music performance.

Three properties make it the appropriate benchmark for this work. First, a subset of takes carries proficiency demonstrator annotations, in which domain experts assign each participant to one of four increasing levels — Novice, Early Expert, Intermediate Expert, Late Expert — based on training history and demonstrated competence. This supplies a four-level ordinal ground truth on the scale the expertise literature argues for, rather than a binary one. Second, the simultaneous ego/exo capture means viewpoint can be treated as a controlled experimental variable rather than a confound: the same performance, the same ground-truth label, two genuinely different views. Third, per-take task_start_sec and task_end_sec annotations mark the interval in which the annotated activity is actually being performed, which makes temporal trimming a controlled ablation rather than a heuristic.

The proficiency labels carry one property that must be stated because it bounds every result in Chapter 4: they are participant-level judgements. A skill level is assigned to a person on the basis of their training history and demonstrated competence, and that single label is then inherited by every take that person recorded. A take in which a participant performs atypically — unusually well, or unusually badly — is invisible to this scheme.

As a cross-domain control this work also uses JIGSAWS (Gao et al., 2014), the JHU–ISI Gesture and Skill Assessment Working Set: a robotic surgical dataset of suturing, knot-tying and needle-passing performed by surgeons of documented experience, annotated with self-reported hours of robotic surgical experience that map to novice/intermediate/expert categories. It is a genuinely out-of-distribution test — a different capture setup, a different domain vocabulary, and a different underlying skill construct.

1.3 Vision-language model architecture

Contemporary vision-language models (VLMs) pair a pre-trained vision encoder with a large language model through a learned projection, following the visual-instruction-tuning recipe popularised by LLaVA (Liu et al., 2023). An image is encoded into a sequence of visual tokens, those tokens are projected into the language model's embedding space, and the language model then attends over them jointly with the text prompt. Video extends this by encoding several frames and concatenating their visual tokens into the same context.

Video-LLaVA (Lin et al., 2024) aligns image and video representations before projection so that a single projection serves both, and uses a Vicuna-7B (Chiang et al., 2023) language backbone with a CLIP ViT-L/14 vision encoder (Radford et al., 2021). The Qwen-VL family (Bai et al., 2025) adopts a native dynamic-resolution vision transformer with window attention and a multimodal rotary positional encoding designed to represent time explicitly; Qwen3-VL extends this to substantially longer video contexts. Proprietary systems such as the Gemini family (Gemini Team, Google, 2023) accept a raw video file directly through an upload API and perform their own internal sampling, so the frame budget is not exposed to the caller at all.

Two architectural facts drive several results in Chapter 4 and are worth stating precisely here.

Frame budget is bounded by context budget. Each frame is encoded as a fixed number of visual tokens. For Video-LLaVA's CLIP ViT-L/14 encoder at 224×224 resolution, that number is 256 tokens per frame. With a Vicuna-7B backbone limited to 4,096 tokens, $16 \times 256 = 4{,}096$ tokens are consumed by the frames alone, before a single word of prompt text is added. Increasing the frame budget is therefore not a free parameter: past a model-specific threshold it does not merely fail to help, it destroys the prompt. Qwen2.5-VL, with a 128,000 token window, is nowhere near this limit at any frame count used here, so the same ablation means different things for the two models.

Sampling paradigm is a genuine experimental variable. Fixed-count extraction gives a 5-second clip and a 90-second clip the same number of frames, and therefore radically different effective frame rates. Duration-proportional sampling holds the rate constant and lets the token count vary with clip length. These are different views of the same video, and conflating them under the single heading "frame count" obscures a real methodological axis. This dissertation treats it as a first-class variable, and groups the four evaluated models accordingly (§3.3).

1.4 Known limitations of vision-language models

A growing body of evidence suggests that VLM visual understanding is shallower than headline benchmark scores imply, and several of these findings anticipate the failure mode documented in Chapter 4.

Contrastively pre-trained vision-language models behave in important respects like bags of words, showing limited sensitivity to word order and to relational composition (Yuksekgonul et al., 2023), and image-text matching fails on minimally contrastive pairs constructed to require compositional reasoning (Thrush et al., 2022). More directly relevant here, "Eyes Wide Shut" (Tong et al., 2024) demonstrates that multimodal LLMs fail on visual details that are trivially discriminable in CLIP feature space itself. The implication is structural: the failure lies downstream of the encoder — in the projection and in the language model — rather than in perception. The localisation experiment of §4.7 is a within-task version of exactly this argument.

Evaluations of temporal understanding are similarly discouraging. TempCompass (Liu et al., 2024) finds that video LLMs frequently answer video questions correctly from a single frame, exploiting static appearance rather than motion, and that benchmark scores degrade far less than they should when the input frames are shuffled — indicating that many models treat video as an unordered set of images rather than as a temporal sequence. Feng et al. (2025) report the same insensitivity to frame order. This matters for skill assessment more than for most video tasks, because the discriminative signal for proficiency is almost entirely dynamic: what distinguishes an expert climber from a novice is the timing and economy of movement, not the appearance of any individual frame.

Separately, instruction-tuned language models are known to carry strong label priors under zero-shot and few-shot classification, and to require calibration before their outputs are usable as classifications (Zhao et al., 2021). What has not previously been quantified, to the author's knowledge, is how completely this prior dominates on a visual evaluative task, and how effectively a class-balanced benchmark disguises that domination as honest chance-level uncertainty.

1.5 Prompting strategies

Because the models are evaluated zero-shot, the prompt is the only lever available. Three strands of the prompting literature inform the four formats defined in §3.4.

Constrained-output prompting. Asking for a single label from a fixed set is the most direct instantiation of a classification task and the easiest to score automatically. Its weakness is that it gives the model nothing to condition on beyond its own prior: there is no intermediate computation between seeing the video and emitting a token. This is precisely the condition under which label priors are expected to dominate.

Chain-of-thought prompting. Wei et al. (2022) showed that prompting a model to produce intermediate reasoning steps before its answer improves performance on multi-step problems, and Kojima et al. (2022) showed that a generic instruction to reason step by step recovers much of the benefit without task-specific exemplars. Applied to skill assessment, the motivation is diagnostic as well as performance-driven: if a model is required to state its observations and the errors it sees before committing to a label, then a run in which the description is accurate but the label is wrong localises the failure at the description-to-label step rather than at perception.

Role and persona framing. Kong et al. (2024) report that assigning a model an expert role, and naming the criteria that role would apply, improves performance on evaluative tasks. This is the most direct available test of the vocabulary hypothesis that motivates this work: if the model fails because it lacks the evaluative vocabulary a coach would use, then supplying that vocabulary explicitly — body alignment, movement fluency, technical precision — should help.

These three strategies are treated in Chapter 3 as parallel interventions rather than as a difficulty hierarchy. The structured prompt supplies a procedure without vocabulary; the reasoning prompt supplies vocabulary without a procedure. Neither is a more advanced version of the other, and the difference in their length is not the experimental variable.

Chapter 2

Methodology

This chapter describes how vision-language models were evaluated on skill-level assessment. The evaluation is designed as an escalating investigation rather than a flat grid of conditions: a feasibility test first, then ablations that remove the obvious input-side explanations one at a time, then diagnostic controls that ask where inside the model the failure occurs. Each stage is justified by what the previous one ruled out, and §3.1 sets out the eight hypotheses that structure that sequence.

One design decision shapes everything that follows: the four models are split into two groups by how they ingest video (§3.3). A model given a fixed eight frames and a model given one frame per second are not seeing the same thing, so comparing their accuracies directly would confound the sampling strategy with the model itself. Frame count is a controlled variable only inside the first group, and the two groups are reported separately throughout Chapter 4.

A second safeguard is needed before any accuracy in this study can be read at face value, and it is defined with the reporting conventions in §4.1: the collapse criterion, which decides whether a result counts as evidence of skill judgement at all. On a class-balanced benchmark a model that answers with the same label every time scores exactly chance, which is indistinguishable from honest uncertainty unless the predicted-label distribution is inspected. Every accuracy reported in this dissertation is therefore paired with per-class recall and the distribution of predicted labels.

The remaining sections cover the benchmarks built from EgoExo4D and JIGSAWS and the activities and camera views they use (§3.2), the four prompt formats and the mechanism each is designed to test (§3.4), and the two evaluation pipelines, generation settings and execution environment (§3.5).

2.1 Approach

The research question is whether a pretrained vision-language model can assess how well a person performs an activity, and, if it cannot, whether the failure lies in perception or in language. The task is operationalised in the simplest form that admits a clean answer: each model receives either the video itself or a set of frames sampled from it, together with a text prompt asking for the performer's skill level, and its answer is scored against the expert-assigned ground truth. Nothing is fine-tuned and no labelled examples appear in the prompt, so the model must map from what it learned in pretraining directly onto the label set.

The central hypothesis is that it cannot make this judgement, and the proposed reason is a linguistic one. Skill is an evaluative property, but the image-text and video-text corpora these models are trained on are overwhelmingly descriptive: captions record what a person is doing, rarely how well they are doing it, and almost never in the vocabulary a coach or examiner would use to grade it. A model may therefore represent the performance accurately while still having no reliable route from that representation to the word "Novice" or "Late Expert".

This hypothesis has a specific consequence for the design of the study, and it is what separates this work from a benchmark report. Establishing that a model scores at chance is straightforward; establishing why requires first eliminating the input-side explanations — too few frames, irrelevant footage, the wrong camera, too small a sample, an unusually hard activity — and then testing the two pathways separately. Hypotheses H1 to H6 do the eliminating; H7 and H8 do the separating, by asking whether the skill signal survives in the visual features without any language generation, and whether it can be recovered from expert language with no video at all. Table 2.1 lists all eight and where each is tested.

Table 2.1: Hypotheses and where each is tested.

#	Hypothesis	Test	Reported in
H1	Vision-language models will not assess human skill level above chance in a zero-shot setting	All models, all prompt formats, both viewpoints, both primary activities	§4.2
H2	The observed failure is primarily label collapse rather than random guessing	Predicted-label distributions and per-class recall, collapse criterion applied uniformly	§4.3
H3	Increasing the number of frames will improve accuracy, because more frames provide more visual evidence	Frame count varied at 8, 16 and 64 frames (Qwen2.5-VL); 8 and 16 (Video-LLaVA)	§4.4.1
H4	Removing idle or off-activity footage by trimming to the annotated task window will improve accuracy	Entire-clip versus trimmed-clip, paired on identical clips	§4.4.2
H5	Exocentric viewpoint will outperform egocentric viewpoint, because it shows whole-body posture directly relevant to skill assessment, whereas egocentric shows mainly hands and the immediate surroundings	Paired ego/exo runs, identical clips, prompts and frame budgets	§4.5.1
H6	Performance will differ meaningfully across activities (climbing, dance, basketball, JIGSAWS, mixed-activity)	Basketball, mixed-activity benchmark, JIGSAWS suturing	§4.6
H7	Skill-discriminative signal is present in visual features even where language output fails	Frozen CLIP features, linear classifier, no language generation	§4.7.2
H8	Skill level is recoverable from expert language alone	Text-only classification from expert commentary, no video	§4.7.3

2.2 Dataset and benchmark construction

All benchmarks are drawn from a multi-view dataset of skilled human activities called EgoExo4D (Grauman et al., 2024). Each recording is referred to as a take that captures one participant performing one activity. Every take is recorded at the same time by a head-mounted egocentric camera and by four static exocentric cameras. A subset of takes is annotated on a four-level increasing proficiency scale: Novice, Early Expert, Intermediate Expert, Late Expert.

Benchmarks were built from the combination of train and validation proficiency annotation files. The dataset contains activities like bouldering, basketball, dance, music, cooking and soccer. Each activity is associated with skill labels, distributed as shown in Table 2.2.

Table 2.2: Proficiency-label distribution by activity in the combined EgoExo4D train and validation annotation files.

Activity	Total	Novice	Early Exp.	Interm. Exp.	Late Exp.
Rock climbing	782	205	212	261	104
Basketball	681	75	118	203	285
Dance	532	41	260	88	143
Music	188	0	14	62	112
Cooking	103	44	33	26	0
Soccer	84	0	0	27	57

This distribution decides which activities can be used and how. Music contains no Novice clips, Cooking contains no Late Expert clips, and Soccer has only two levels. Hence none of these can be used to construct a balanced benchmark, but they were selected for the mixed-activity benchmark. A balanced four-class benchmark can only be constructed for rock climbing, basketball and dance.

Bouldering is the primary activity because it has the most balanced distribution of clips across all four proficiency levels. It also has a clear task structure: a climber either completes a route or falls. The quality of the climb can be assessed based on factors such as reach, foot placement and hip position. Dance is selected as the second domain because it provides a strong contrast to climbing across several dimensions. It involves whole-body movement rather than being mainly focused on the upper body, and does not have a clear success or failure outcome. Therefore, if the same model fails in both domains, it is unlikely to be caused by features specific to climbing.

Four additional activity sets were used to evaluate the generalisability of the findings beyond the primary domain. Basketball was selected from EgoExo4D and has the same proficiency label schema. The mixed benchmark combines basketball, music, cooking and soccer while excluding climbing and dance. The JHU–ISI Gesture and Skill Assessment Working Set (JIGSAWS) (Gao et al., 2014), a surgical activity dataset, was used as a cross-domain evaluation set. It consists of robotic suturing videos with expert-assigned skill labels.

Table 2.3: Benchmark composition and chance baselines.

Benchmark	Clips	Composition	Chance baseline
Climbing binary	50	25 Novice, 25 Late Expert	50%
Climbing four-class	100	25 per level	25%
Dance binary	50	25 Novice, 25 Late Expert	50%
Dance four-class	100	25 per level	25%
Mixed-activity four-class	96	Basketball 46, Cooking 21, Music 19, Soccer 10	25%
Basketball binary	34	12 Novice, 22 Late Expert	50%; majority 64.7%
JIGSAWS suturing binary	29	19 Novice, 10 Expert	50%; majority 65.5%
Three-class climbing	30	10 per class	33.3%

In the binary task, Novice and Late Expert are used as the two classes because they represent the extreme ends of the proficiency scale. Late Expert clips are scored as correct when the model answers "Expert". The intermediate levels are entirely excluded, as this maximises visual distinction between the two classes. If a model cannot tell between the two extremes, then further analysis of the four-class task is less informative.

2.2.1 Camera viewpoints

An egocentric view is recorded from a camera worn by the participant and provides a first-person perspective of the activity. An exocentric view is captured by a static camera placed at a distance from the participant performing the activity. Each EgoExo4D take provides five synchronised video files: one egocentric view from the Aria glasses (aria01) and four exocentric views from static cameras (cam01 through cam04). Every clip was evaluated twice, once from each view, using the same prompt, fixed frame count and the same generation settings. The ground-truth label is identical for both views because they are frame-aligned recordings of the same performance. Therefore, camera viewpoint is the only factor associated with a difference in model output.

The two views carry different information. The egocentric view shows the hold the performer is reaching for, their own hands entering the frame, and the climbing wall at close range. The whole-body structure, hip position and leg extension that a coach would use are not visible at all. The exocentric view shows the full body but from a distance, so fine detail such as grip type is harder to resolve. Both viewpoints were included in the evaluation because each provides different visual information.

2.3 Models evaluated

Four models were selected for evaluation. They were selected based on the size of the language backbone's context window and the mechanism for processing videos into the model.

Table 2.4: Models evaluated, with language backbone, context window and video-input mechanism. Group 1 receives a fixed number of extracted frames; Group 2 receives frames at a duration-proportional rate.

Model	Params	Language backbone	Context	Video input	Grp.
Qwen2.5-VL-7B-Instruct	7B	Qwen2.5	128,000	Extracted frames	1
Video-LLaVA-7B-hf	7B	Vicuna-7B-v1.5	4,096	Extracted frames	1
Qwen3-VL-8B-Instruct	8B	Qwen3	262,144	Frames at ∼1 fps (substitute)	2
Gemini 3.1 Flash-Lite	n/d	Not disclosed	Large	Native, ∼1 fps (API)	2

All four models do not process the video in the same way. Some sample a fixed number of frames regardless of the clip duration, whereas others sample frames according to the length of the video. This difference affects the amount of visual information available to each model for any given clip. This makes direct accuracy comparisons across the two approaches difficult to interpret. Differences in performance may therefore result from the frame-sampling strategy rather than from differences in the models' underlying capabilities. For this reason, these models are evaluated in two groups.

Group 1: fixed-count frame extraction (Qwen2.5-VL, Video-LLaVA)

A fixed number of frames (8 or 16) is uniformly sampled per clip regardless of its duration. This means that a 30-second clip and a 90-second clip both receive exactly eight frames. This keeps the visual token budget consistent across clips, which makes frame count a controlled variable. However, longer clips are represented with lower temporal resolution.

Qwen2.5-VL-7B has a context window of 128,000 tokens, which is large enough that none of the prompt lengths or frame counts used in this study reach the model's capacity limit. This means any change in its output as frame count changes reflects the extra visual information, not the model running low on context space. In contrast, Video-LLaVA-7B uses a Vicuna-7B-v1.5 backbone with a fixed context window of 4,096 tokens. Its CLIP ViT-L/14 encoder generates 256 visual tokens per frame at a resolution of 224×224 (Radford et al., 2021). Consequently, processing 16 frames alone occupies the entire context window before the prompt is added. For this reason, Video-LLaVA is evaluated at 8 frames only in the headline comparisons; the 16-frame runs are retained and reported in §4.9 as evidence of the failure mode itself.

Group 2: duration-proportional native sampling (Gemini 3.1 Flash-Lite, Qwen3-VL)

Video is sampled at approximately one frame per second internally. This means a longer clip generates a proportionally higher number of frames. The climbing benchmark contains clips ranging from 3.5 to 82 seconds in duration, with an average length of approximately 31 seconds. This gives roughly 3 to 82 frames per clip at this rate.

Gemini processes the raw video file through the API's upload mechanism and performs sampling internally. Qwen3-VL was initially designed to follow the same approach, with its video processor requested to extract $n = \text{round}(d)$ frames, where d is the clip duration in seconds. This would have allowed the two models to be directly comparable; however, it was unsuccessful. Due to a fallback issue in the processor, the requested frame count was ignored and videos kept being sampled at 24 fps. The higher sampling rate resulted in excessive GPU memory usage and repeated out-of-memory errors. The Qwen3-VL results reported in this study were therefore derived using a manual substitution method implemented with OpenCV. This approach extracts $n = \max(1, \text{round}(d))$ frames per clip. It provides the model with a list of images instead of native video input. This difference is consistently maintained across all reported results: the figures for Gemini represent actual native video processing, while those for Qwen3-VL correspond to frame extraction designed to approximate the same sampling rate.

2.4 Prompt design

Four prompt types were used across all models and activities. Each prompt is designed to test a different aspect of skill assessment.

BINARY (chance 50%).

> Is this person a Novice or an Expert at this activity? Answer only: Novice or Expert

FOURCLASS (chance 25%).

> What is the skill level of the person in this video? Answer only one:
> Novice / Early Expert / Intermediate Expert / Late Expert

STRUCTURED (chance 25%).

> Watch these frames carefully.
> Step 1: Describe the person's body position and technique in detail.
> Step 2: Identify any errors or imprecisions in their movement.
> Step 3: Classify skill level as exactly one of: Novice / Early Expert / Intermediate Expert / Late Expert.
> Format your answer as:
> Observations: ...
> Errors: ...
> Skill Level: ...

REASONING (chance 25%).

> You are an expert coach evaluating this person's technique.
> Focus on: body alignment, movement fluency, technical precision.
> What is their skill level? Novice / Early Expert / Intermediate Expert / Late Expert
> Explain your reasoning.

Table 2.5: The four prompt formats and the mechanism each is designed to test.

Prompt	Labels	Chance	Mechanism under test
Binary	2	50%	Whether skill can be separated at all when only two options exist
Fourclass	4	25%	Whether finer granularity can be resolved on the dataset's own label set
Structured	4	25%	Whether forcing explicit observation and error statement before the verdict improves it
Reasoning	4	25%	Whether an expert persona with named criteria improves the verdict, with no forced observation step

The binary and four-class prompts are used as fundamental baselines for comparison with the longer prompts.

The structured prompt implements chain-of-thought prompting (Wei et al., 2022; Kojima et al., 2022) by requiring the model to provide a description and an error analysis before generating a label. This approach separates two potential failure points: first, the model creating an accurate description but assigning an incorrect label; second, the failure lying in translating the observation into the label rather than in the observation itself. In contrast, the reasoning prompt applies role or persona framing (Kong et al., 2024) by presenting the model as an expert evaluator. It explicitly specifies three evaluation dimensions: body alignment, movement fluency and technical precision. Unlike the structured prompt, it does not require the model to provide an observation or error analysis before assigning the final label. This makes it the most direct available test of whether the absence of evaluative vocabulary affects performance.

The four prompts are designed as parallel interventions rather than a hierarchy of difficulty. Although the structured prompt is longer than the reasoning prompt, its length is not the experimental variable. Instead, the structured prompt offers a procedural framework without vocabulary, while the reasoning prompt offers vocabulary without procedure. Neither represents a more advanced version of the other.

2.5 Implementation

Two separate evaluation pipelines were developed, one for each paradigm described in §3.3. Both pipelines process the same benchmark JSON files, apply the same prompts, and produce outputs formatted according to the same CSV schema. This approach ensures comparability of results within groups and allows for shared analysis scripts. Each benchmark entry includes clip_id, take_folder, video_path_ego, video_path_exo, activity and ground_truth, with ground-truth strings precisely matching the label set.

2.5.1 Fixed-count frame extraction (Group 1)

The frames were extracted using OpenCV (Bradski, 2000). Models in this group receive a fixed number of frames n per clip, sampled uniformly across the clip regardless of its duration. For a clip of N total frames, the sampled indices are $\lfloor kN/n \rfloor$ for $k = 0 \dots n-1$. The script then seeks each specified index and decodes the corresponding frame. Extracted frames are resized to 420×360 pixels and converted from BGR to RGB. They were converted into PIL images before being processed by the model's own image processor, which performs the model-specific normalisation and patching. Resizing before processing maintains consistent memory usage over extended runs and preserves the aspect ratio sufficiently to avoid subject distortion.

Two fixed frame counts were evaluated: 8 and 16. The 8-frame condition was used as the baseline. It was applied across every model, activity, prompt and view wherever the model's context window allowed. The 16-frame condition was introduced to test Hypothesis 3. Doubling the frame count rather than incrementing it was chosen deliberately: a significant difference should be visible if frame count influences performance, whereas a smaller increment could be hidden by noise. A 64-frame condition was added subsequently for Qwen2.5-VL, whose context window permits it, extending the ablation to eight times the baseline budget.

The motivation for varying frame count originated from unresolved issues in the video-language domain. TempCompass (Liu et al., 2024) and Feng et al. (2025) report that video-LLM benchmark scores decline only slightly when the input frames are shuffled. This indicates that many models process video inputs as unordered image sets rather than temporal sequences. This raises the question of whether increasing frame count improves performance, which this ablation aims to address.

2.5.2 Entire clip versus trimmed clip

Two sampling strategies were implemented for video scope. The entire-clip condition sampled frames uniformly across the full clip duration. In contrast, the trimmed-clip condition restricted sampling to the interval defined by task_start_sec and task_end_sec from the EgoExo4D takes.json annotations. This interval represents the period during which the annotated activity is actually being performed.

The purpose of this comparison was to determine whether uniform sampling across the full take introduces irrelevant visual content. For example, a climbing take usually begins with the participant standing at the base or performing calibration gestures. Instead of assuming that these periods represent significant wasted input, a “dead time” was calculated for each clip: the duration outside the annotated activity window. These results are reported in §4.4.2 alongside frame inspections. The trimmed condition tests Hypothesis 4 by assessing whether excluding irrelevant portions affects outcomes.

2.5.3 Native video input (Group 2)

Gemini was accessed via the google-genai SDK. Videos were uploaded using the Files API, after which the prompt was provided together with the returned file handle. The model internally samples roughly one frame per second. Trimmed inputs were produced externally using FFmpeg before upload, because the API exposes no frame-indexing mechanism.

Qwen3-VL was initially designed to use the same native-video approach locally, but this led to a significant implementation issue. When fps=1.0 was specified without the required metadata, the loader silently defaulted to fps=24. This increased the frame count by approximately 24 times and caused GPU memory exhaustion. Explicitly requesting metadata with return_video_metadata=True triggers an AttributeError due to a version mismatch between qwen_vl_utils and the installed version of transformers. Both issues manifested as out-of-memory failures rather than as configuration errors, which made the root cause difficult to diagnose.

The chosen workaround avoided the native-video path. Clip duration was extracted using OpenCV and the number of frames was calculated as $\mathsf{round}(d)$ for a clip of duration d seconds. The corresponding frames were then given as a list of images. This approximates one-frame-per-second sampling without triggering the faulty metadata process, and was confirmed on individual clips (for example, a 9.2-second clip yields nine frames). This approach mitigates but does not fully resolve memory pressure during full benchmark runs.

2.5.4 Generation settings

All text generation employed greedy decoding without sampling to ensure deterministic outputs. As a result, variation between experimental conditions cannot be explained by random sampling during decoding. The maximum number of newly generated tokens was set to 50 for binary and four-class prompts, because expected answers were one or two words. For structured and reasoning prompts it was extended to 300–400 tokens, because they require detailed descriptions and reasoning. Model weights were loaded using either float16 or bfloat16 precision.

2.5.5 Execution environment

Experiments ran on the University of Edinburgh's Informatics computing cluster under SLURM, using NVIDIA H200 and A6000 GPUs. Models were accessed via the HuggingFace transformers library. Results were written incrementally to CSV files to improve the reliability of long batch jobs, enabling resumption after interruption. The GPU cache was cleared after each clip. Gemini runs applied the same resume strategy to manage daily API quotas.

Chapter 3

Results and Discussion

3.1 Reporting conventions

There are four conventions that must be defined before presenting the results. Each of these affects the interpretation of the numerical results reported in this chapter. Three of them represent methodological decisions rather than merely alternative ways of presenting the results.

First, distinct chance baselines are defined for each task type. The binary task comprises 50 clips with a chance accuracy of 50%, while the three four-class tasks (fourclass, structured, reasoning) each use 100 clips with a 25% chance baseline. These baselines are not directly comparable and are therefore presented separately. When both task types need to be shown together, accuracies are plotted relative to their own chance baselines.

Second, accuracy alone is insufficient to analyse model behaviour. All accuracy results reported in this chapter are accompanied by per-class recall and the distribution of predicted labels. This is necessary because the main problem in the experiments is single-label collapse, which can be hidden when only accuracy is reported. For example, a model in the JIGSAWS task that consistently labelled every clip “Novice” achieved 65.5% accuracy due to class imbalance, which misleadingly indicates strong performance.

Third, collapse is defined as a condition in which only one ground-truth class has non-zero recall. This indicates that the model effectively generates a single label across the benchmark. Collapsed conditions are reported rather than discarded, because they are findings about model behaviour.

Lastly, binary conditions were evaluated using a one-sample binomial test against a probability of 0.5. For four-class conditions, a chi-squared goodness-of-fit test compared the predicted-label distribution to a uniform distribution. It is important to clarify that the four-class p-values do not indicate accuracy above chance, but rather measure deviation from uniformity. For example, a model labelling all clips as Novice achieves 25% accuracy, matching chance, yet yields a highly significant p-value ($p < 10^{-40}$) precisely because the distribution is fully non-uniform. Four-class test results should therefore be interpreted alongside assessments of label diversity, and not as standalone evidence of model performance.

3.2 Can these models assess skill at all?

Hypothesis (H1). Vision-language models will not assess human skill level above chance in a zero-shot setting.

Test. Each of the four models was evaluated on the binary and four-class benchmarks for both primary activities, across every prompt format, viewpoint and frame count.

Verdict. Across approximately 130 executed conditions, only one achieves above-chance accuracy together with a balanced prediction distribution: the binary result for a single activity, viewpoint and model.

3.2.1 Binary skill discrimination

The binary task represents the minimum feasibility test. If a model cannot distinguish a novice from an expert when given only two options and the two ends of the scale, attempting finer-grained distinctions is difficult to justify.

Table 3.1: Binary results, climbing (Novice against Expert, balanced $n = 50$, chance $= 50\%$).

Model	Input	Trim	View	Acc.	Nov. rec.	Exp. rec.	Predicted	Verdict
Gemini	native	entire	ego	76.0%	84%	68%	Nov 29, Exp 21	Genuine signal, $p = 0.0003$
Gemini	native	entire	exo	60.0%	52%	68%	Exp 29, Nov 21	Not significant ($p = 0.26$)
Video-LLaVA	16 fr	entire	exo	76.0%	88%	64%	mixed	Confounded artefact, §4.9
Video-LLaVA	16 fr	entire	ego	32.0%	48%	16%	mixed	Below chance, same clips
Video-LLaVA	8 fr	trimmed	exo	60.0%	100%	20%	Nov 45, Exp 5	Weak, near-collapse
Qwen3-VL	∼1 fps	entire	exo	54.0%	100%	8%	Nov 48, Exp 2	Near-collapse
Qwen3-VL	∼1 fps	entire	ego	50.0%	100%	0%	Nov 50	Collapse
Qwen2.5-VL	8/16/64 fr	all	all	50.0%	100%	0%	Nov 50	Total collapse, $p = 1.0$

Table 3.2: Binary results, dance (balanced $n = 50$, chance $= 50\%$).

Model	Input	View	Acc.	Nov. rec.	Exp. rec.	Predicted	Verdict
Gemini	native	exo	60.0%	52%	68%	Exp 29, Nov 21	Not significant ($p = 0.20$)
Gemini	native	ego	52.0%	100%	4%	Nov 49, Exp 1	Near-collapse
Qwen3-VL	∼1 fps	exo	50.0%	92%	8%	Nov 46, Exp 4	Near-collapse
Qwen3-VL	∼1 fps	ego	50.0%	100%	0%	Nov 50	Collapse
Qwen2.5-VL	all 8 cond.	both	50.0%	100%	0%	Nov 50	Collapse in every run
Video-LLaVA	all 8 cond.	both	50.0%	100%	0%	Nov 50	Collapse in every run

Accuracy is simply the midpoint of the two recalls because the binary test set is balanced. A model that repeatedly outputs one label therefore achieves exactly 50% accuracy, and Figure 3.1 makes this visible directly: every condition whose two bars sum to 100 is scoring exactly chance, no matter how high the taller bar is.

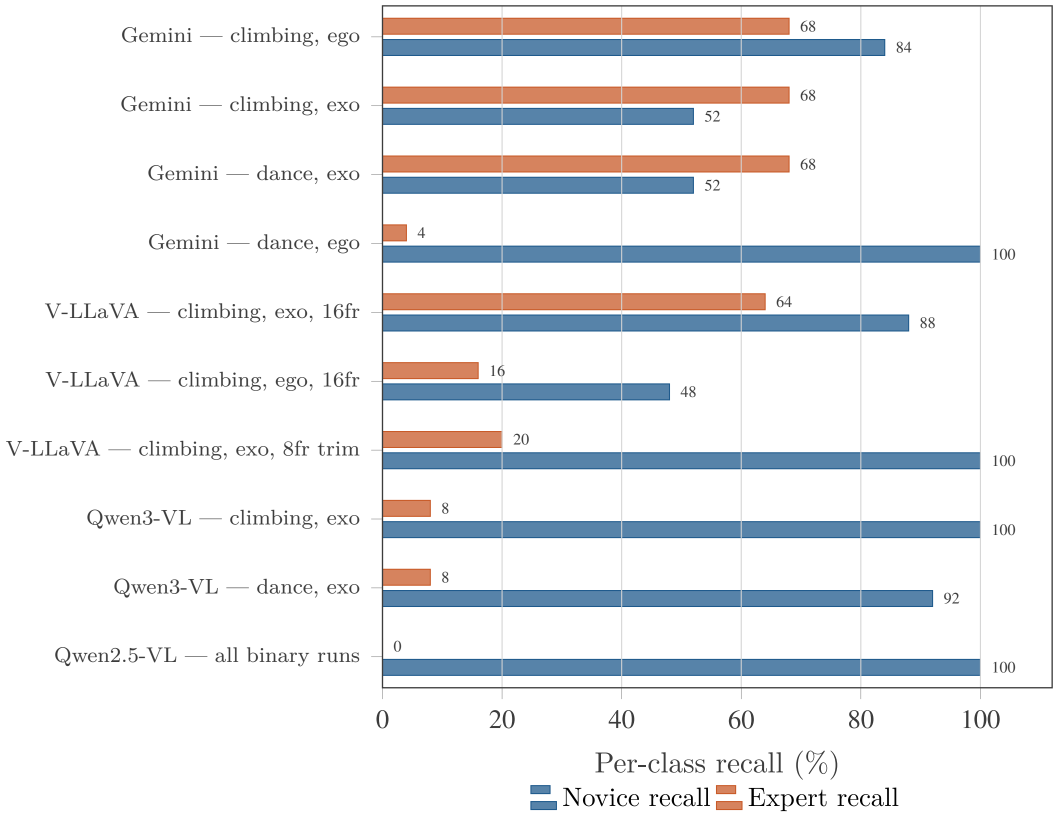

Figure 3.1: Per-class recall on the binary task. Because the binary benchmark is balanced 25/25, accuracy is the mean of the two bars; any condition whose bars sum to 100 scores exactly 50%, i.e. chance. Only Gemini on egocentric climbing (84% Novice, 68% Expert) has both bars substantially above 50%, which is what a genuine discrimination looks like. Qwen2.5-VL sits at the degenerate extreme, recalling every Novice and no Expert in every single binary run.

The results highlight three key observations.

Gemini's egocentric climbing result is the only statistically significant above-chance performance observed. Its 76% accuracy combines 84% Novice recall, 68% Expert recall and a nearly balanced prediction split of 29 to 21. This indicates that the strongest tested model can detect the simplest proficiency distinction. Section 4.5.1 shows that this result does not generalise across different viewpoints or activities.

Qwen2.5-VL consistently predicted "Novice" for every clip. This happened in all tested conditions, including varied frame counts (8 to 64), full and trimmed clips, egocentric and exocentric views, climbing and dance activities, and different cameras. Its 50% accuracy reflects the class-balanced base rate rather than any assessment ability.

Video-LLaVA's 76% accuracy is not comparable to Gemini's performance. For the same clips it scored 32% on the egocentric view, which is below chance. Section 4.9 clarifies that the 16-frame binary prompt exceeded the model's context window, causing silent truncation. This result therefore cannot be considered a valid evaluation of the model's capability and is reported only with its limitation noted.

3.2.2 Four-class skill discrimination

The four-class results are consistent across models. No model shows a reliable above-chance result. After re-scoring, pooled accuracies are 29% for Gemini on climbing and 25% on dance, while Qwen3-VL achieves 29% and 27% respectively. Qwen2.5-VL (24% climbing, 28% dance) and Video-LLaVA (25% on both) fall in the same band. Every model, on every activity, sits within roughly four points of the 25% chance baseline.

Table 3.3: Climbing, representative four-class results ($n = 100$, chance $= 25\%$).

Model	Prompt	Input	View	Acc.	Recall (N/EE/IE/LE)	Note
Qwen3-VL	fourclass	∼1 fps	ego	37%	92 / 0 / 56 / 0	Best four-class climbing result of any model
Gemini	structured	native	exo	33%	40 / 0 / 92 / 0	Gemini's best climbing result
Gemini	fourclass	native	ego	32%	48 / 0 / 80 / 0	
Video-LLaVA	structured	8 fr	ego, trim	30%	60 / 4 / 16 / 40	Only run with substantial Late Expert recall
Qwen2.5-VL	fourclass	16 fr	exo	27%	92 / 0 / 16 / 0	
Qwen3-VL	structured	∼1 fps	exo	27%	24 / 0 / 64 / 20	First climbing run to predict Late Expert
Qwen2.5-VL	reasoning	16 fr	exo	18%	24 / 16 / 32 / 0	Only climbing run resolving three classes at once

Table 3.4: Dance, representative four-class results ($n = 100$, chance = 25%).

Model	Prompt	Input	View	Acc.	Recall (N/EE/IE/LE)	Note
Qwen2.5-VL	structured	16 fr	exo	41%	88 / 0 / 76 / 0	Best Qwen2.5-VL result of any kind
Qwen2.5-VL	structured	16 fr	exo, trim	39%	68 / 0 / 88 / 0	Close second
Qwen2.5-VL	fourclass	16 fr	exo, trim	34%	64 / 0 / 72 / 0	
Gemini	reasoning	native	exo	32%	28 / 4 / 96 / 0	Gemini's best dance result
Video-LLaVA	reasoning	8 fr	ego, trim	31%	0 / 16 / 12 / 96	Late-Expert-dominated, see §4.3
Qwen3-VL	structured	∼1 fps	exo	30%	12 / 4 / 80 / 24	Best Qwen3-VL dance result

The per-class results reveal a different pattern of model performance. The "Early Expert" class is almost never correctly identified by any model, with recall of zero in nearly every run. The only exception is the re-parsed Qwen2.5-VL reasoning run at 16%, which remains below chance at 18% overall accuracy. Similarly, the "Late Expert" class shows minimal improvement. Gemini on climbing never predicts either Expert class in any four-class, structured or reasoning run, instead limiting its predictions to Novice and Intermediate Expert.

The strongest result is 41% accuracy on dance, achieved by Qwen2.5-VL using the structured prompt with 16 frames from an exocentric viewpoint. It has per-class recalls of 88% Novice, 0% Early Expert, 76% Intermediate Expert and 0% Late Expert. Similarly, Qwen3-VL achieves 37% on climbing with per-class recalls of 92 / 0 / 56 / 0. Both models predict only two of the four labels, scoring zero on the others.

Figure 3.2 makes the underlying independence explicit. It shows, for each true skill level, the distribution of labels Gemini actually predicts on climbing, pooled over all three four-way prompts. If the model were discriminating, the four bars would differ markedly from row to row. Instead they are close to identical.

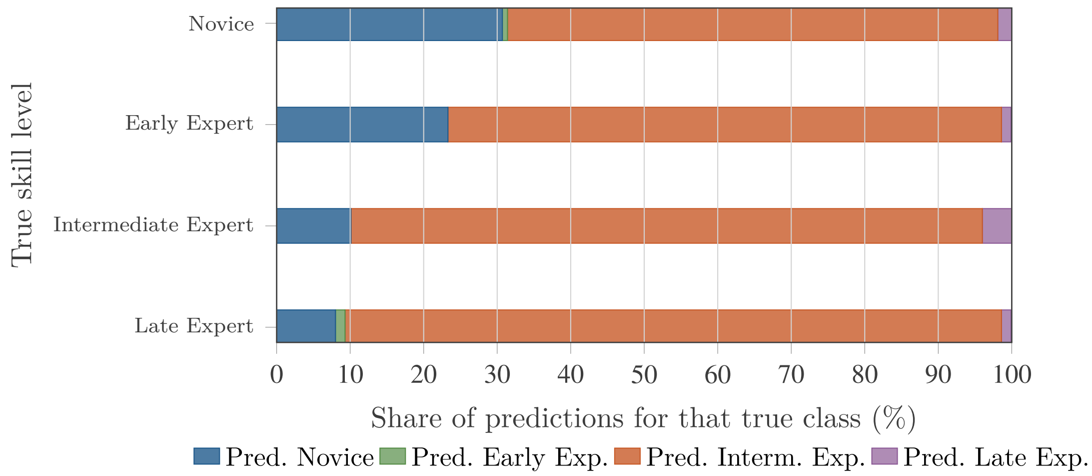

Figure 3.2: Predicted-label distribution by true class: Gemini, climbing, four-way prompts pooled ($n \approx 150$ valid answers per row). Each bar is one row of a row-normalised confusion matrix. The rows are nearly identical, which is to say the predicted distribution barely responds to the true skill level: Gemini labels 86–89% of Intermediate and Late Expert climbers "Intermediate Expert" — and 67% of true novices the same way. Early Expert is essentially never predicted at all.

3.3 The failure is label collapse, not noise

Hypothesis (H2). The observed failure is primarily label collapse rather than random guessing.

Test. Predicted-label distributions and per-class accuracies were analysed across all experimental conditions, using the uniform collapse criterion defined in §4.1.

Verdict. H2 is supported. The models do not randomly guess. Instead, they respond in the same way regardless of the input. On a balanced benchmark, this constant answer produces chance-level accuracy.

Approximately 43% of the tested conditions show total or near-total label collapse, with the model assigning a single label to between 87% and 100% of clips. Table 3.5 breaks this down by model. Qwen2.5-VL predicts "Novice" for essentially all 50 clips in every binary condition, consistently scoring 50.0%. Chi-squared goodness-of-fit tests produce p-values below 10^{-20} in the most extreme cases. Collapse is therefore not a marginal effect but the dominant pattern in the experiment.

Table 3.5: Prevalence of label collapse by model. A condition is one model $\times$ activity $\times$ prompt $\times$ view $\times$ frame count $\times$ trim cell. Totals are sensitive to how jointly reported ego and exo cells are individuated and should be read as close approximations.

Model	Conditions	Collapsed	% collapsed
Video-LLaVA-7B	49	28	57.1%
Qwen2.5-VL-7B	89	42	47.2%
Qwen3-VL-8B	18	4	22.2%
Gemini	16	0	0.0%
All models	172	74	43.0%

Figure 3.3 shows the same phenomenon on a continuous scale rather than a binary criterion, by reporting the share of clips receiving whichever label the model most often produced. Two observations follow.

Robustness to label collapse differs significantly by model, and does not depend only on model size. Gemini never collapses in any of its 16 conditions, while Video-LLaVA collapses in more than half of its 49. Video-LLaVA-7B collapses more than Qwen2.5-VL-7B despite having the same parameter count, which is due in part to its far more limited context capacity.

This tendency should not, however, be mistaken for capability. Although Gemini collapses in zero conditions, it still underperforms on the four-class task: its best climbing result is 33%, with 0% recall on both Early and Late Expert.

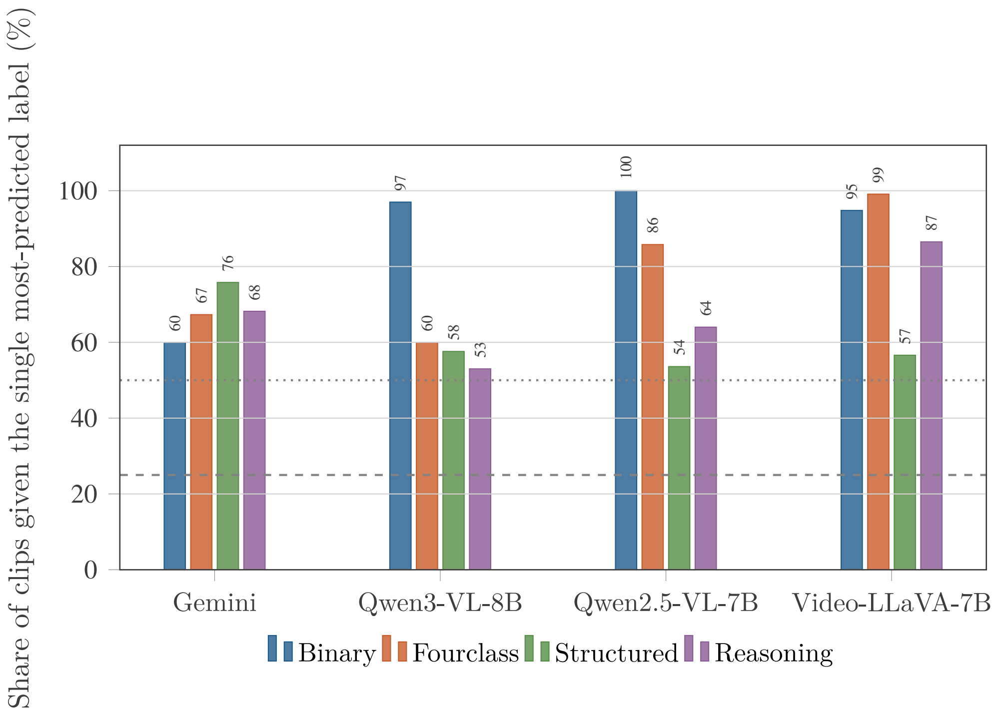

Figure 3.3: Label-collapse severity across models, activities and prompt formats. Each bar reports the share of clips receiving the most-predicted label, averaged over climbing and dance. A model spreading its answers evenly would sit on the dashed line (25%, four-label prompts) or the dotted line (50%, binary). Bars approaching 100% mean one answer for nearly every clip. Binary is the most collapse-prone format for three of the four models, and Qwen2.5-VL reaches a full 100% on it.

3.4 The visual input is not the cause

Sections 4.2 and 4.3 show that the failure is consistent rather than random. A natural explanation is that the models are not receiving enough of the relevant visual content. Two ablations test this directly.

3.4.1 Frame budget

Hypothesis (H3). Increasing the number of frames will improve accuracy, because more frames provide more visual evidence.

Test. Qwen2.5-VL was evaluated with 8 and 16 frames, with an additional 64-frame condition. Video-LLaVA was tested at 8 and 16 frames. This comparison is specific to Group 1, because the duration-proportional models were never evaluated at a fixed frame count (§3.3).

Verdict. H3 is not supported. Increasing the number of frames does not improve performance.

For Qwen2.5-VL, increasing the frame count from 8 to 16 does not meaningfully affect four-class accuracy, and the binary result remains a 100% Novice collapse at both counts. The 64-frame condition, at eight times the standard budget, does not improve binary performance either; four-class accuracy stays between 20% and 32%. The one notable exception is climbing's reasoning-exocentric condition, which improves from 18% at both 8 and 16 frames to 32% at 64 frames. On dance, moving to 64 frames produced results that were flat or worse than at 8 or 16 frames: the highest overall accuracy of 41%, achieved by the structured-exocentric condition at 16 frames, drops to 27% at 64. The frame-count effect is therefore specific to a particular activity and prompt, rather than a general benefit of supplying more visual information. For Video-LLaVA, increasing the frame count failed outright for architectural reasons detailed in §4.9.

It is also worth noting how the one improvement happened. At 8 and 16 frames the climbing reasoning-exo run is Novice-leaning; at 64 frames Novice is never predicted at all and the distribution becomes Early/Intermediate-dominant. More frames changed what the model defaults to, not whether it defaults.

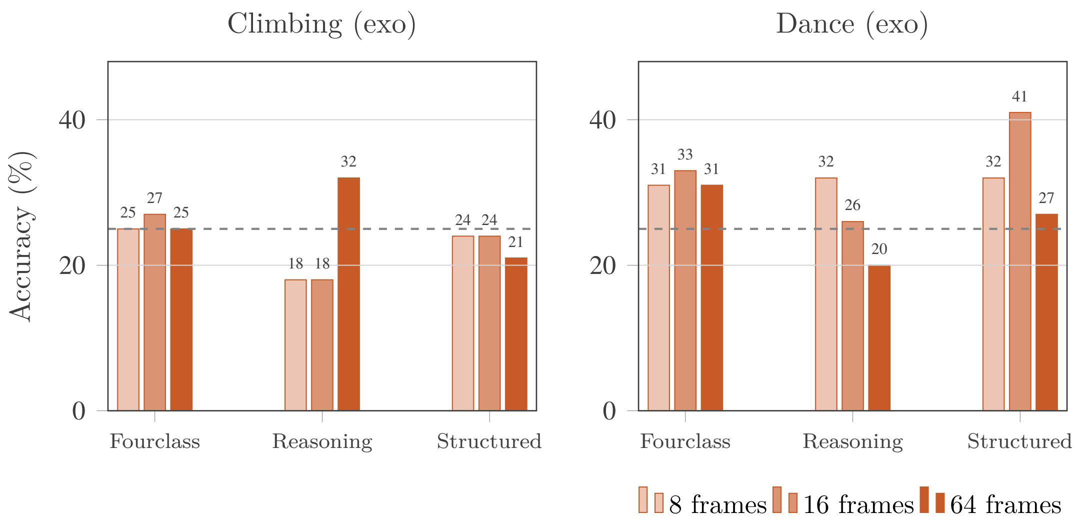

Figure 3.4: Effect of sampled frame count (Qwen2.5-VL-7B, entire clip, exocentric view). The dashed line is the 25% four-class chance baseline. Eight-fold more frames produces no systematic gain: on climbing only the reasoning prompt moves (18% → 32%), and on dance every prompt is flat or worse at 64 frames, including the project's best Qwen2.5-VL result (41% → 27%). The one improvement does not replicate across activities, so it is not evidence that more visual evidence helps.

3.4.2 Temporal trimming

Hypothesis (H4). Removing idle or off-activity footage by trimming to the annotated task window will improve accuracy.

Test. Every Group 1 condition was run twice, once on the entire clip and once on the clip trimmed to the annotated task_start_sec–task_end_sec window.

Verdict. H4 is not supported. Trimming improves accuracy about as often as it reduces it.

When the model is provided with individual frames from a known Late Expert clip, it answers "Novice" for a person who is standing or looking at a route, but scores higher for frames of active climbing. Uniform sampling across a full clip includes mostly frames other than peak execution, which motivated the trimming experiment.

Qwen2.5-VL and Video-LLaVA binary runs remain almost 100% Novice-collapsed after trimming in every condition. Figure 3.5 shows that the difference varies by up to 13 points between paired ego and exo views of the same clips. The Video-LLaVA climbing structured condition drops from 28.6% to 15.3% on the exo view, while the Qwen2.5-VL dance structured result shows the same pattern at a smaller scale, dropping from 41% to 39% at 16 frames. These findings indicate that the Novice bias persists despite the removal of the idle frames assumed to cause it.

The effect of trimming is unpredictable, because it also produced notable single-view improvements. Video-LLaVA's climbing structured result increased by 5.3 points in the egocentric view, and its dance structured result by 7.0 points in the exocentric view. However, these do not represent a consistent overall effect: the climbing egocentric improvement was not matched in the exocentric view of the same clips, which instead decreased by 13.3 points. A genuine reduction in input noise should not help one camera and hurt another on identical footage.

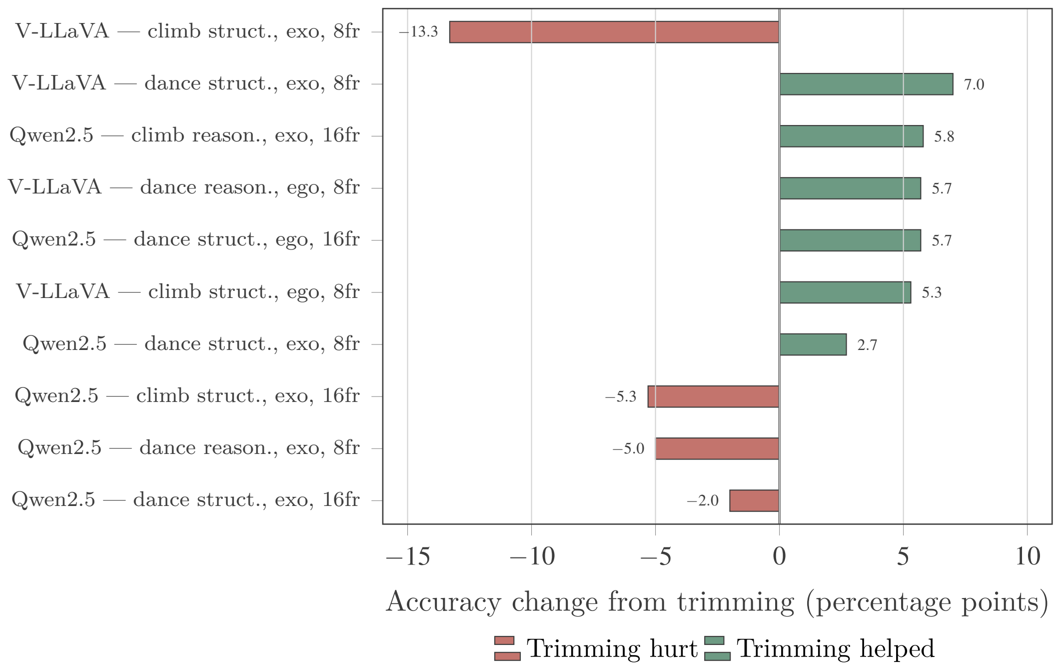

Figure 3.5: Effect of trimming to the annotated task window: the ten largest changes. Bars to the right of zero are conditions where trimming helped, bars to the left where it hurt. The distribution is close to symmetric, which is what "no effect" looks like on a noisy measurement. The decisive pair is Video-LLaVA climbing structured: the same clips trimmed the same way gain 5.3 points on the egocentric camera and lose 13.3 on the exocentric one. Removing genuinely irrelevant footage should not do that.

3.5 Viewpoint and benchmark scale

3.5.1 Viewpoint

Hypothesis (H5). Exocentric viewpoint will outperform egocentric viewpoint, because it shows whole-body posture directly relevant to skill assessment, whereas egocentric shows mainly hands and the immediate surroundings.

Test. Both viewpoint runs were paired on identical clips, prompts and frame budgets.

Verdict. H5 is not supported. Exocentric viewpoint does not consistently outperform egocentric viewpoint; the direction depends on both the model and the activity.

Gemini's results contradict the hypothesis outright. It achieved 76% binary accuracy on egocentric climbing clips compared with 60% on exocentric. The same model on dance shows the opposite pattern, with exocentric outperforming egocentric (60% versus 52%). A fixed mechanism of the kind the hypothesis proposes cannot explain an activity-dependent reversal.

Qwen2.5-VL shows a different pattern. Egocentric input strengthens the model's Novice default rather than reducing accuracy through lost detail. Novice predictions increase from 75% to 85% on climbing and from 36% to 90% on dance under egocentric framing. Every reliable Qwen2.5-VL result in this chapter comes from an exocentric view. This does not indicate that exocentric input is inherently more informative; rather, it appears to be less prone to collapse. For instance, exocentric accuracy on the dance structured task is 41%, but the corresponding egocentric result collapses to exactly 25%. The finding matches the direction predicted by the hypothesis but not the mechanism behind it: the exocentric advantage is not that it shows more of the body, but that the egocentric view is more likely to trigger collapse.

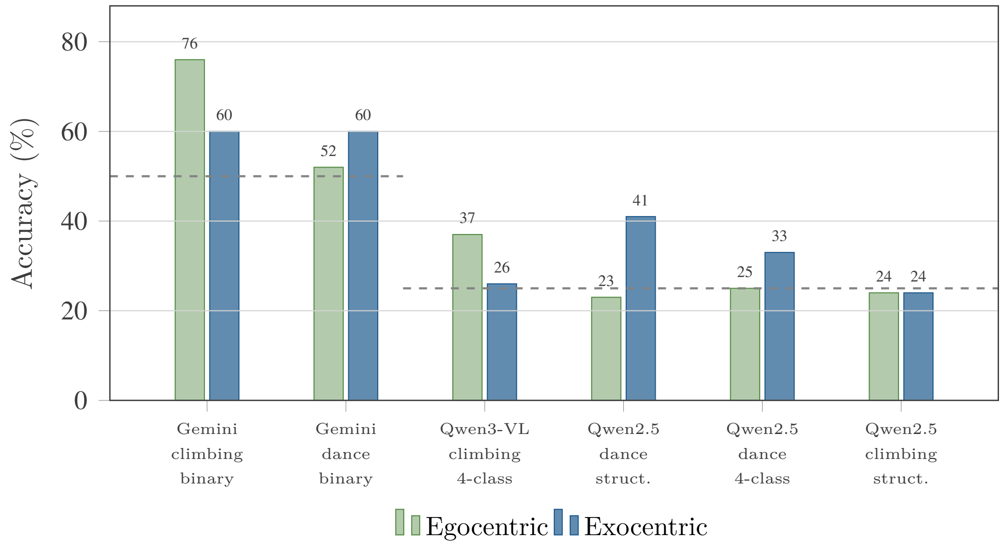

Figure 3.6: Effect of camera viewpoint on paired runs (identical clips, prompts and frame budgets). The dashed lines mark the relevant chance baseline — 50% for the two binary conditions on the left, 25% for the four-class conditions on the right. The direction of the viewpoint effect reverses between models and between activities: Gemini prefers ego on climbing and exo on dance, while Qwen2.5-VL's exocentric advantage on dance is an escape from collapse rather than a gain from better body visibility. No single viewpoint is reliably better.

3.5.2 Benchmark scale

A final check excludes a small-sample artefact. The benchmark was rebuilt at approximately 100 clips per class ($n = 389$, with Late Expert capped at 89 by data availability) and re-run at 8 frames.

Table 3.6: Small-sample against large-sample comparison, Qwen2.5-VL-7B, climbing, 8 frames, entire clip.

Prompt	View	$n = 100$	$n = 389$	Delta
Reasoning	exo	18.0%	16.7%	−1.3
Structured	exo	24.0%	23.4%	−0.6
Reasoning	ego	24.0% (collapsed)	25.4% (collapsed)	+1.4
Structured	ego	19.0%	25.4%	+6.4

The same pattern holds at roughly four times the sample size. Egocentric views still collapse and exocentric views partially resolve. Three of the four conditions fall within 1.4 points of the original figures. The reasoning-egocentric condition remains collapsed, predicting Novice for 382 of 389 clips. These findings are not a sampling artefact.

3.6 Generalisation across activity domains

The remaining plausible explanation is that bouldering is especially hard to judge, or that its annotations are unusually noisy.

Hypothesis (H6). Performance will differ meaningfully across activities (climbing, dance, basketball, JIGSAWS, mixed-activity).

Test. The evaluation was extended to basketball from EgoExo4D, to a mixed benchmark pooling basketball, music, cooking and soccer with climbing and dance excluded by design, and to JIGSAWS robotic suturing as an out-of-distribution dataset recorded with a single close-range camera and requiring a domain-specific rephrased prompt.

Verdict. H6 is not supported. Every domain tested showed the collapse.

Table 3.7: Cross-domain and cross-dataset results, Qwen2.5-VL-7B, 8 frames.

Test	Format	View	Acc.	Chance	Distribution	Reading
Basketball (EgoExo4D)	binary	ego+exo	24.0%	50%	Novice 34, Unknown 16	Below chance ($p = 0.0003$); 16/50 rows unparseable from missing video, 15 on Expert-labelled clips
JIGSAWS suturing	binary	exo	65.5%	50%	Novice 29/29	Not signal; majority-class artefact of a 19 N / 10 E imbalance; $p = 0.136$
Mixed activity	binary	ego+exo	25.0%	50%	Novice 100/100	Collapsed; far below chance
Mixed activity	structured	exo	19.0%	25%	IE 51, N 44, Unk 5	Kept; below chance
Mixed activity	structured	ego	23.0%	25%	IE 48, Unk 27, N 25	Kept; at chance

The failure holds across seven distinct activity types — bouldering, dance, basketball, surgical suturing, music, cooking and soccer — spanning two datasets.

3.6.1 JIGSAWS

At 65.5%, JIGSAWS is the highest cross-domain score in the study, and the only one above chance. But it was produced by a model that answered "Novice" 29 times out of 29. Per-class accuracy is 100% on Novice and 0% on Expert. The result is not statistically significant: a binomial test gives $p = 0.136$. If the benchmark had contained an equal mix of classes, this same behaviour would have scored 50%; had it been mostly Expert clips, it would have scored 34.5%. The accuracy score reflects how the benchmark is balanced, not how well the model is performing. This is the single clearest justification for the reporting convention of §4.1.

3.6.2 A mixed-activity benchmark

The mixed-activity structured runs show uneven results across activities: basketball and cooking retain more signal than music and soccer.

Table 3.8: Per-activity accuracy within the mixed benchmark, structured prompt, 8 frames, trimmed.

Activity	n	Correct (exo)	Correct (ego)	Combined rate
Basketball	48	12	9	21.9%
Cooking	21	5	8	31.0%
Music (piano)	20	2	5	17.5%
Soccer	11	0	1	4.5%

One possible explanation is that whatever weak signal remains tends to align with activities involving clear, visible interactions with objects — a ball being handled, a knife being used — where success or failure can be inferred from a single frame. This contrasts with activities relying on whole-body motion or precise motor timing. The idea is consistent with the diagnosis in §4.7. It must, however, be treated as speculative: sample sizes range from 11 to 48, the differences are untested for significance, and all accuracies are at or below chance, so the comparison is between two kinds of failure rather than between success and failure. It is offered as a hypothesis for future work, not as a finding.

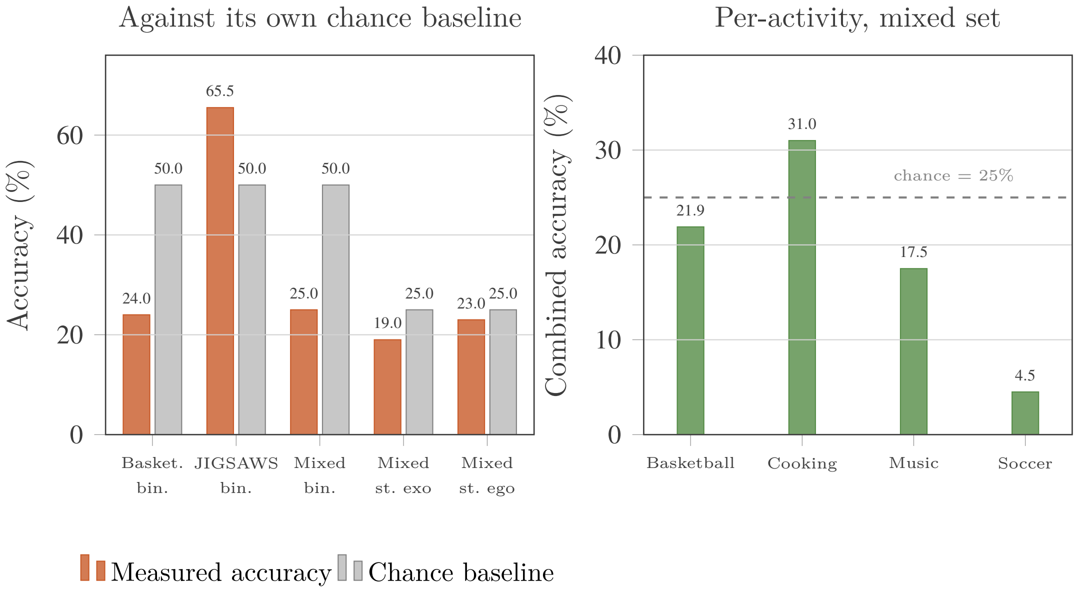

Figure 3.7: Cross-domain generalisation (Qwen2.5-VL-7B, 8 frames). Left: every cross-domain test plotted against its own chance baseline. The only bar clearing its baseline is JIGSAWS, and that is the majority-class artefact of §4.6.1 — a model that answered "Novice" 29 times out of 29 on a 19/10 imbalanced set. Right: the per-activity breakdown inside the mixed benchmark. Only cooking exceeds chance, on 21 clips; soccer collapses almost completely. The comparison is between two kinds of failure, not between success and failure.

3.7 Locating the failure: perception or language?

Sections 4.4 to 4.6 exclude the input-side explanations one by one: it is not the frame budget, not idle footage, not the camera, not the sample size, and not the activity. What remains is to ask where inside the model the failure occurs. Two controls answer this, and they are the most informative experiments in the project because neither of them requires the model to succeed at the task at all.

3.7.1 Perception is intact

A scene-content diagnostic first confirms that the models see the video. Asked to describe what is present rather than to judge it, all four models correctly identify the activity, the number of people, the equipment and the dominant colours, on clips whose skill judgement they then get wrong. Whatever is failing, it is not video decoding and it is not gross perception.

3.7.2 Vision alone succeeds (H7)

Hypothesis (H7). Skill-discriminative signal is present in visual features even where language output fails.

Test. A linear classifier was trained on frozen CLIP features extracted from the same exocentric climbing clips, with no language generation anywhere in the pipeline, and evaluated by 5-fold cross-validation.

Verdict. H7 is supported, decisively.

The linear probe reaches 67.0% four-class accuracy (5-fold cross-validation; 95% CI 62.5–71.5%) on clips where the VLMs score 24–25%. The features are frozen: nothing was learned about vision, only a linear partition of a representation the encoder already produced. The discriminative signal for proficiency is therefore present, and linearly accessible, in the visual representation before it reaches any language model. It is also worth noting that this exceeds the 47–53% reported by trained, task-specific supervised methods on the same four-class benchmark, which places the result well outside the range that could be attributed to noise.

3.7.3 Language alone also fails (H8)

Hypothesis (H8). Skill level is recoverable from expert language alone.

Test. The models were given an expert's written coaching commentary describing a climbing attempt — no video, no frames, nothing visual — and asked for the same four-class judgement.

Verdict. H8 is not supported. Text-only performance lands in the same band as video.

Qwen2.5-VL scores 26.0% and Video-LLaVA 33.0% on the text-only four-class task, against video-based bests of roughly 27% and 30% respectively for the same models on climbing. Text does not fail where video succeeds, nor does it clearly outperform video. Both text-only runs also reproduce the same collapse asymmetry seen throughout: Qwen2.5-VL virtually never predicts Novice from text (3 of 100), exactly as it never does from egocentric video, while Video-LLaVA over-predicts it (35 of 100), mirroring its video-side bias.

3.7.4 What the two controls jointly establish

Taken together the three results localise the failure precisely. Perception is intact. A linear read-out of the visual representation classifies skill at 67%. Supplying the evidence in explicit expert language instead of pixels does not help either. The bottleneck is therefore neither the visual encoder nor the availability of the evidence: it is the step at which either modality must be converted into the correct categorical label. This is a within-task instance of the argument made by Tong et al. (2024), and it is the central claim of this dissertation.

3.8 Qualitative analysis: what the models actually say

Accuracy tables record what the models answered; this section analyses how they answered, and identifies a distinct failure signature for each model family.

Qwen2.5-VL under a free-text prompt frequently hedges rather than committing to a single class, writing the literal disjunction "novice or early expert". This occurred 24 times in a 16-frame trimmed climbing run, 12 times in the corresponding entire-clip run, and 11 and 7 times in the 8-frame versions, against effectively zero instances under four-class or structured prompts. The behaviour suggests a failure to make a confident distinction rather than calibrated uncertainty, since it concentrates on one specific class pair and disappears entirely when the output format forces a commitment. A representative answer, on a clip whose ground truth is Novice, reads: the climber appears to be a novice or early expert based on the following observations … the climber's technique and body alignment suggest that they are at a novice or early expert level. The re-parser scores such answers as Unknown rather than resolving the disjunction arbitrarily, since any Early Expert credit awarded from one would be a coin flip.

Fixed-frame models also respond stereotypically. Qwen2.5-VL opens 99.8% of its egocentric dance reasoning answers with the same sentence, and 86.6% of its egocentric climbing answers, while Video-LLaVA reuses identical phrasing across unrelated clips. Such repetition indicates that the response is being generated from a prior rather than from the video content, which weakens the evidential value of the text considerably.

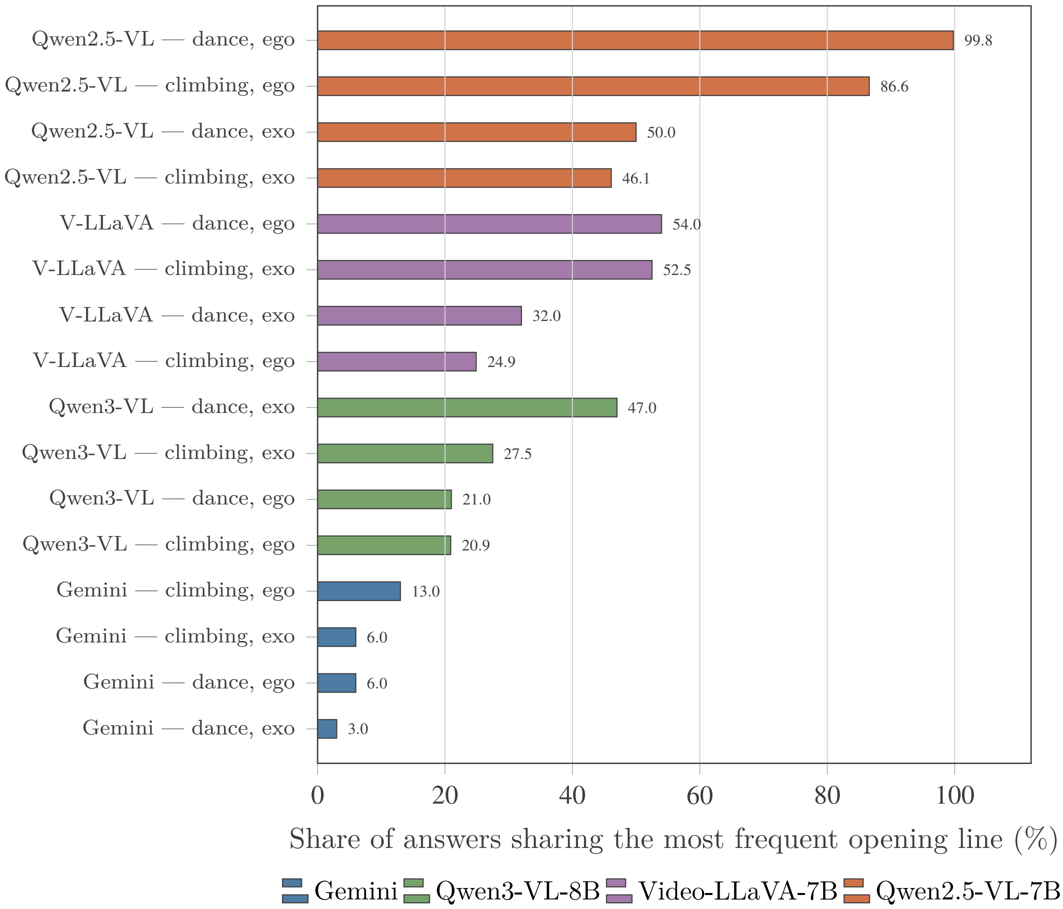

Figure 3.8: Stereotypy of reasoning responses: the share of answers in each condition beginning with the single most frequent opening line. A model writing genuinely per-clip text would sit near the left edge. The fixed-frame models do not: Qwen2.5-VL opens 99.8% of its egocentric dance answers identically, and 86.6% of its egocentric climbing answers. Gemini sits at 3–13% throughout, writing distinct prose for almost every clip — and still, per Figure 3.2, produces judgements nearly independent of the ground truth. Specificity of language is not evidence of grounding.

By contrast, the duration-proportional models produce descriptive, clip-specific analyses running to 1,000–2,500 characters, naming particular holds, movements and errors; at most 3–13% of Gemini's answers share an opening line. Despite this verbosity, the final judgement remains poorly correlated with the ground truth, as Figure 3.2 showed. This is arguably the more dangerous of the two failure modes, because fluent, specific, plausible prose is exactly what a human reader uses as evidence that a model understood the video.

Occasionally Gemini refuses to answer when the video does not contain the requested content — for example when the camera wearer is filming other dancers rather than dancing, or when a team is talking in a briefing rather than climbing. Although these refusals are scored as incorrect, they are arguably the most genuinely grounded visual behaviour observed anywhere in the project, and they expose real noise in the egocentric ground truth, which is systematically less reliable than the exocentric annotations.

A consistent pattern holds across models: the extreme classes are not recognised. Recall for Early Expert is zero in most four-class conditions, and Late Expert recall is very low except for Qwen3-VL, which shows modest improvement. The four-level ordinal scale is effectively answered as a two-level one, corroborating the collapse analysis of §4.3.

3.9 Reliability and limitations

Several constraints bound the results above and are reported here rather than distributed through the chapter.

Video-LLaVA's context ceiling. The 16-frame runs are not valid evaluations of the model. At 256 visual tokens per frame, $16 \times 256 = 4{,}096$ tokens fill the entire Vicuna-7B context window before any prompt text is added. The structured and reasoning scripts abort generation on an explicit guard and emit no usable text; the four-class script has no such guard and silently proceeds with an over-length prompt, producing garbage parsed as Unknown for 83–95 of 100 rows. The 76% binary result of Table 3.1 was produced under this silent truncation, which is why it is reported as a confounded artefact rather than as a capability. The below-chance 32% egocentric result on the same clips is the corroborating evidence.

Missing source files. Nine of ten uniandes_bouldering takes referenced by the climbing benchmark have no video on disk for either the exocentric or the egocentric stream, imposing a systemic error floor of roughly 9% on every locally executed climbing run. Sixteen of fifty original basketball clips were similarly absent, and that benchmark was rebuilt from the annotation pool; 15 of those 16 unparseable rows fall on Expert-labelled clips specifically, which is why the basketball binary result sits below chance. Gemini was unaffected, because it reads uploaded video rather than the local mirror — which confounds direct model comparison on exactly those clips.

Qwen3-VL is a substitute pipeline. As described in §3.5.3, all reported Qwen3-VL numbers come from OpenCV frame extraction at a matched ∼1 fps rate, not from the model's native video path. Gemini and Qwen3-VL are therefore comparable in sampling rate, not in ingestion mechanism.

Parser sensitivity. An early keyword parser misread negated and aspirational phrasing ("not yet a Late Expert" scored as Late Expert) and resolved hedged disjunctions by whichever label it happened to check first. All numbers in this chapter come from a negation-aware re-parser that drops label mentions preceded by a negation cue and scores unresolved disjunctions as Unknown. Re-parsing moved individual conditions by up to 5 points and changed one Video-LLaVA run's collapse status, but altered no qualitative conclusion.

Run-to-run non-determinism. Despite greedy decoding, two independent Qwen2.5-VL runs on identical data disagreed on up to 86 of 100 predictions. Single-run point estimates therefore carry more variance than is conventionally assumed, which is a further reason to read the collapse pattern rather than any individual accuracy figure.

Scope of the CLIP probe. The localisation control of §4.7.2, while decisive, was run on exocentric climbing only.

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